Food and Drug Administration CFR Title 21 Part 11

FDA CFR Title 21 overview

The Code of Federal Regulations (CFR) contains the rules and regulations for executive departments and agencies of the US federal government. Each of the 50 titles of the CFR addresses a different regulated area.

FDA CFR Title 21 regulates food and drugs manufactured or consumed in the United States, under the jurisdiction of the Food and Drug Administration (FDA), the Drug Enforcement Administration, and the Office of National Drug Control Policy. The regulations outlined in CFR Title 21 Part 11 set the ground rules for the technology systems that manage information used by organizations subject to FDA oversight. Any technology system that governs such GxP processes as Good Laboratory Practices (GLP), Good Clinical Practices (GCP), and Good Manufacturing Practices (GMP) also requires validation of its adherence to GxP.

CFR Title 21 Part 11 sets requirements to ensure that electronic records and signatures are trustworthy, reliable, and equivalent substitutes for paper records and handwritten signatures. It also offers guidelines to improve the security of computer systems in FDA-regulated industries. Subject companies must prove that their processes and products work as they're designed to, and if these process and products change, they must revalidate that proof. The best practices guidelines cover:

- Standard operating procedures and controls that support electronic records and signatures such as data backup, security, and computer system validation.
- Features that ensure that the computer system is secure, contains audit trails for data values, and ensures the integrity of electronic signatures.
- Validation and documentation that supply evidence that the system does what is intended, and that users can detect when the system isn't working as designed.

Microsoft and FDA CFR Title 21

Microsoft enterprise cloud services undergo regular independent third-party SOC 1 Type 2 and SOC 2 Type 2 audits and are certified according to ISO/IEC 27001 and ISO/IEC 27018 standards.

Although these regular audits and certifications don't specifically focus on FDA regulatory compliance, their purpose and objectives are similar in nature to those of CFR Title 21 Part 11, and serve to help ensure the confidentiality, integrity, and availability of data stored in Microsoft

cloud services. Our qualification approach is also based on industry best practices, including the International Society for Pharmaceutical Engineering (ISPE) GAMP series of Good Practices Guides and the Pharmaceutical Inspection Cooperation Scheme (PIC/S) Good Practices for Computerized Systems in Regulated GxP Environments.

Customers can request access to the compliance reports, subject to nondisclosure agreement terms and conditions, through their Microsoft account representative, or through the Service Trust Portal .

Microsoft in-scope cloud platforms and services

Although no certification exists for complying with CFR Title 21 Part 11, the following Microsoft enterprise cloud services undergo independent, third-party audits. These audits can help you with your compliance efforts. These services include:

- Azure: Cloud Services, Storage, Traffic Manager, Virtual Machines, and Virtual Network
- Azure DevOps
- Microsoft Intune
- Microsoft Dynamics 365 and Dynamics 365 U.S. Government
- Office 365 and Office 365 U.S. Government

Audits, reports, and certificates

The audit reports for SOC 1 and SOC 2 Type 2, ISO/IEC 27001, and ISO/IEC 27018 standards attest to the effectiveness of the controls Microsoft implements. These reports can help you comply with FDA CFR Title 21 Part 11.

Frequently asked questions

To whom does the standard apply?

FDA CFR Title 21 Part 11 applies to organizations with products and services that deal in FDA-regulated aspects of the research, clinical study, maintenance, manufacturing, and distribution of life science products.

How do Microsoft enterprise cloud services demonstrate compliance with FDA CFR Title 21 Part 11?

Using the formal audits prepared by third parties for SOC 1 Type 2, SOC 2 Type 2, ISO/IEC 27001, and ISO/IEC 27018, Microsoft shows how relevant controls noted within these reports address the requirements.

Audited controls that Microsoft implements help ensure the confidentiality, integrity, and availability of data. These controls correspond to the applicable regulatory requirements defined in Title 21 Part 11 that Microsoft is responsible for. The qualification guidelines for Azure and Office 365 detail how Microsoft audit controls correspond to those requirements.

How can I get copies of the auditor's reports?

The Service Trust Portal provides independently audited compliance reports. You can use the portal to request audit reports so that your auditors can compare Microsoft's cloud services results with your own legal and regulatory requirements.

Can I use Microsoft's compliance in the certification process for my organization?

Yes. The independent third-party compliance reports of the IEC/ISO 27001, ISO/IEC 27018, SOC 1, and SOC 2 standards attest to the effectiveness of Microsoft controls. Microsoft enterprise cloud customers can use the audited controls described in these related reports as part of their own CFR Title 21 Part 11 risk analysis and qualification efforts. Customers who build and deploy applications subject to FDA regulation are responsible for ensuring that their applications meet FDA requirements.

What are Microsoft's responsibilities for maintaining compliance with this standard?

Microsoft ensures that its enterprise cloud services meet the terms defined within the governing Online Services Terms and applicable Service Level Agreements (SLAs). These terms define our responsibility for implementing and maintaining controls adequate to secure and monitor the system.

Use Microsoft Purview Compliance Manager to assess your risk

Microsoft Purview Compliance Manager is a feature in the Microsoft Purview portal to help you understand your organization's compliance posture and take actions to help reduce risks. Compliance Manager offers a premium template for building an assessment for this regulation. Find the template in the **assessment templates** page in Compliance Manager. Learn how to build assessments in Compliance Manager.

Resources

- Azure - GxP (FDA 21 CFR Part 11)
- Code of Federal Regulations Title 21
- FDA guidance for industry Part 11: Electronic records and signatures
- Microsoft Online Services Terms

- Microsoft Cloud for Government
- Microsoft Trust Center
- Compliance on the Microsoft Trust Center

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